

NeuroRAG: Polychronous Multimodal Episodic Memory

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Abstract

Modern enterprise information systems contain heterogeneous streams of information that differ fundamentally in representation, temporal structure, and semantics. Relational databases encode structured attributes and transactions; knowledge graphs encode entities and relations; textual systems encode sequential symbolic information; telemetry and event streams additionally encode temporal dynamics. Conventional memory systems typically transform these modalities into a common vector space and subsequently perform similarity search, graph traversal, or learned retrieval. Such approaches provide effective retrieval but do not naturally model one-shot associative learning, temporal binding, pattern completion, continual online learning, or biologically inspired memory consolidation.

This work proposes a Polychronous Multimodal Episodic Memory (PMEM) architecture based on a spiking neural substrate inspired by the entorhinal cortex (EC), dentate gyrus (DG), hippocampal CA3, and CA1. Heterogeneous modalities are independently transformed into temporally structured spike trains and converged into a common entorhinal representation. The DG performs sparse pattern separation, while recurrent CA3 dynamics form autoassociative attractor states capable of completing memories from partial or cross-modal cues. Spike-timing-dependent plasticity (STDP), neuromodulatory gating, synaptic delays, and polychronous convergence provide mechanisms for rapid associative binding. CA1 provides a learned readout from the latent hippocampal state back to modality-specific representations. Subsequent extensions introduce episodic temporal sequences, replay-based consolidation, continual learning, homeostatic plasticity, modular scaling, and cross-modal retrieval.

The resulting system is not intended to replace relational databases, graph databases, or language models. Instead, it defines a neural associative memory substrate operating over these systems, in which structured records, graph relations, and textual episodes become temporally encoded neural events. The central research hypothesis is that multimodal information can be bound through spike timing and recurrent dynamics without requiring a single shared semantic embedding or an explicit retrieval index. The current prototype demonstrates encoder reconstruction, CA3 pattern completion, multimodal binding, engram stability, novelty-sensitive plasticity, and continuous two-stage encoding; however, several components remain prototype-level, particularly pure neural retrieval without symbolic assistance, large-scale capacity, semantic text encoding, continual consolidation, and rigorous comparison against modern memory baselines

1 Introduction

The human brain effortlessly encodes, consolidates, and retrieves memories that bind together disparate sensory modalities—a conversation overheard in a crowded room, the face of a colleague, the context of a meeting room—into coherent episodic traces. Replicating this capability in artificial systems remains one of the central challenges in neuromorphic computing and multimodal machine learning. While deep learning has produced powerful unimodal encoders for vision, language, and structured data, these representations typically remain siloed within modality-specific pipelines, lacking the biological mechanisms that enable cross-modal association, continual learning, and robust retrieval under partial cues.

This paper presents a polychronous multi-modal spiking memory architecture that implements a biologically inspired encoding–consolidation–retrieval cycle using spiking neural networks (SNNs). Our system maps structured relational data, graph-structured knowledge, and natural language text into a unified latent space via modality-specific encoders, and routes these representations through an entorhinal–dentate–hippocampal pathway (EC–DG–CA3–CA1) modeled after the mammalian medial

temporal lobe. Key features of the architecture include: (1) *sparse polychronous coding* in the dentate gyrus (DG) via k -winner-take-all competition; (2) *recurrent attractor dynamics* in CA3 with spike-timing-dependent plasticity (STDP) and homeostatic regulation; (3) *neuromodulated consolidation* gated by a novelty-detection mechanism in the entorhinal cortex (EC); and (4) *cross-modal readout* through a learned CA1 back-projection that enables retrieval of any modality from a partial cue in another.

Unlike conventional memory-augmented neural networks that rely on dense differentiable attention over external key-value stores, our model operates on explicit spike times and delayed synaptic transmission. This temporal structure provides a natural substrate for encoding relational order (e.g., “manages” vs. “reports to”) through polychronous activation patterns, and for distinguishing familiar from novel episodes via energy-based novelty detection. We evaluate the system on a multi-modal episode stream comprising SQL records, knowledge-graph edges, and textual descriptions, measuring retention under interference, separation between similar episodes, continuity across temporally adjacent events, and robustness to modality dropout. Our results demonstrate that the spiking substrate achieves competitive cross-modal retrieval accuracy while exhibiting graceful degradation under partial cues and continual learning without catastrophic forgetting.

2 Related Work

2.1 Spiking Neural Networks and Neuromorphic Memory

Spiking neural networks offer a computational substrate that is temporally precise and energetically efficient, making them attractive candidates for online memory systems. The leaky integrate-and-fire (LIF) model, augmented with synaptic delays and STDP, has been widely used to study polychronous groups—sets of neurons that fire in reproducible temporal patterns despite not being synaptically connected in a feed-forward chain [1]. Recent neuromorphic implementations have scaled these ideas to multi-chip systems for event-based sensory processing [2], yet few have addressed the problem of *multimodal* binding and long-term episodic storage. Our work extends the polychronous framework by introducing structured cross-modal encoders and a consolidation protocol that stabilizes attractor basins through replay, mirroring hippocampal sharp-wave ripple reactivation observed during sleep and quiet wakefulness [3].

2.2 Multimodal Representation Learning

The dominant paradigm for multimodal learning relies on large-scale contrastive pre-training to align vision, language, and audio into a shared embedding space [4]. While effective for zero-shot transfer, these representations are typically static: once computed, they do not adapt to new episodes, nor do they provide mechanisms for partial-cue retrieval or temporal sequencing. Complementary work on multimodal transformers has introduced cross-attention layers that fuse modalities at the token level [5], but such models suffer from quadratic complexity and lack explicit memory structures. Our approach differs by using modality-specific sparse codes that are dynamically bound through a common spiking substrate, enabling retrieval of one modality from a cue in another without requiring joint pre-training of all encoders.

2.3 Episodic Memory and Consolidation in Artificial Systems

Episodic memory architectures in deep learning have drawn inspiration from the hippocampal indexing theory [6], leading to models such as differentiable neural computers (DNCs) [7] and memory networks [8]. These systems use external memory matrices with content-based addressing, updated via gradient descent. While powerful, they do not capture the *temporal* and *sparsity* constraints of biological memory formation. More recent work has explored replay-based consolidation to mitigate catastrophic forgetting [9], but typically replays raw inputs or latent vectors rather than compressed spiking engrams. Our model introduces a two-stage encoding protocol—pre-activation followed by neuromodulated plasticity—that mirrors the transition from encoding to consolidation in the hippocampus, and employs offline replay over DG assemblies to stabilize CA3 attractors without requiring generative replay of raw sensory data.

2.4 Memory-Augmented Neural Networks

Memory-augmented neural networks (MANNs) enhance parametric models with non-parametric external stores, enabling few-shot learning and long-horizon credit assignment [10]. Key-value attention mechanisms allow these models to retrieve relevant past experiences, but the retrieval operation is dense and differentiable, lacking the sharp thresholding and sparse activation characteristic of biological recall. Our CA3 attractor network implements a form of “nearest neighbor” retrieval through recurrent dynamics: partial cues drive the network into a basin of attraction corresponding to a stored episode, with the basin shape shaped by STDP and homeostasis. This provides a mechanistic account of how sparse, spiking representations can support associative memory without attention-based content lookup.

2.5 Cross-Modal Retrieval and Association

Cross-modal retrieval—the ability to retrieve an image given a text description, or vice versa—has been extensively studied in the information retrieval literature [11]. Most approaches learn a joint metric space in which distances reflect semantic similarity. However, these systems typically assume batch training and static corpora. Our architecture supports *online* cross-modal association: as episodes are encoded, the CA1 readout learns to map CA3 engrams back to the support vectors of each modality. We introduce a cross-episode retrieval accuracy (CERA) metric that quantifies how well a cue in one modality retrieves the correct content in another, and demonstrate that the spiking pathway achieves robust CERA even when cues are partial and the memory system is under continual update.

3 Methodology

We describe a polychronous multi-modal spiking memory architecture that implements a biologically inspired encoding–consolidation–retrieval cycle. The system maps structured relational data, graph-structured knowledge, and natural language into a unified spiking substrate via modality-specific encoders, and routes these representations through an entorhinal–dentate–hippocampal (EC–DG–CA3–CA1) pathway. All continuous-time dynamics are simulated with an Euler timestep $\Delta t = 0.1$ ms.

3.1 Problem Formulation

An episode ξ is a tuple (q, g, z) comprising:

- a structured record $q \in \mathcal{Q}$ (e.g., an SQL row with fields {age, salary, city, role, dept});
- a graph edge $g = (u, r, v) \in \mathcal{G}$ with source node u , relation r , and target node v ;
- an optional natural-language utterance $z \in \mathcal{Z}$.

The system processes a stream $\mathcal{X} = \{\xi_1, \xi_2, \dots\}$ and must:

1. **Encode** each episode into a sparse, high-dimensional spiking engram $E_n \subset \{0, \dots, N_{\text{CA3}} - 1\}$.
2. **Consolidate** engrams via neuromodulated STDP and offline replay.
3. **Retrieve** the full episode from a partial cue in any modality, returning reconstructions $\hat{q}, \hat{g}, \hat{z}$ and a relation prediction $\hat{r}$.

3.2 Neural Primitives

3.2.1 Leaky Integrate-and-Fire (LIF) Neuron

Each neuron follows the continuous dynamics

$$\tau_m \frac{dv}{dt} = -(v - v_{\text{rest}}) + I_{\text{syn}}(t), \quad (1)$$

with discrete-time Euler update

$$v_{t+\Delta t} = v_t + \frac{-(v_t - v_{\text{rest}}) + I_{\text{syn}}(t)}{\tau_m} \Delta t. \quad (2)$$

A spike is emitted when $v_t \geq v_{\text{th}} + b_i$, where b_i is a homeostatic offset. After spiking, the membrane resets to v_{reset} and enters an absolute refractory period τ_{ref} . Table 1 lists the default parameters.

Table 1: LIF neuron parameters.

Parameter	Excitatory (RS)	Inhibitory (FS)
τ_m	20 ms	15 ms
v_{rest}	-70 mV	-70 mV
v_{th}	-55 mV	-50 mV
v_{reset}	-75 mV	-75 mV
τ_{ref}	2 ms	1 ms
Δt	0.1 ms	0.1 ms

3.2.2 Delayed Synapse with Exponential PSC

Each synapse carries a weight w , delay d , and synaptic time constant $\tau_{\text{syn}} = 3$ ms. The postsynaptic current is computed from an exponential trace $s(t)$:

$$s(t + \Delta t) = s(t) \exp\left(-\frac{\Delta t}{\tau_{\text{syn}}}\right) + \sum_k w_k \mathbf{1}[t_k + d_k \leq t], \quad (3)$$

$$I_{\text{syn}}(t) = g \cdot s(t), \quad (4)$$

where g is a gain factor and $\mathbf{1}[\cdot]$ is the indicator function. Weight bounds are $w \in [w_{\text{min}}, w_{\text{max}}]$ with defaults $w_{\text{min}} = 0$, $w_{\text{max}} = 10$.

3.2.3 Spike-Timing-Dependent Plasticity (STDP)

When the post-synaptic neuron fires at t_{post} , the delay-adjusted time difference is $\Delta t = t_{\text{post}} - (t_{\text{pre}} + d)$. The weight update is

$$\Delta w = \begin{cases} M A_+ \exp\left(-\frac{\Delta t}{\tau_+}\right), & \Delta t \geq 0, \\ -M A_- \exp\left(\frac{\Delta t}{\tau_-}\right), & \Delta t < 0, \end{cases} \quad (5)$$

with $A_+ = 0.5$, $A_- = 0.48$, $\tau_+ = \tau_- = 12$ ms. The neuromodulatory gain $M \in \{0, 1, 2\}$ gates plasticity: $M = 0$ disables learning, $M = 1$ enables normal plasticity, and $M = 2$ boosts it for novel episodes.

3.3 Modality-Specific Encoders

3.3.1 SQL Encoder: Structured Population Rate Coding

The SQL encoder uses $5 \times 20 = 100$ neurons, one population per field. For a numeric field x , neuron i with center c_i and width σ fires at rate

$$r_i(x) = r_{\text{base}} \exp\left(-\frac{(x - c_i)^2}{2\sigma^2}\right). \quad (6)$$

Over an encoding window $T = 10$ ms, the expected spike count is $\lambda_i = r_i(x) \cdot T / 1000$, which is rounded and clipped to $[0, N_{\text{max}}]$ with $N_{\text{max}} = 5$. Spike times are sampled uniformly in $[0, T]$.

Age. Centers span $[20, 80]$ with $\sigma = 8$ and $r_{\text{base}} = 240$ Hz.

Salary (two-band encoding). To handle both common enterprise salaries and extreme outliers, salary uses two bands:

- *Low band* (10 neurons): linear centers in $[0, 10^5]$, $\sigma = 15000$, $r_{\text{base}} = 280$ Hz.

- *High band* (10 neurons): log-transformed centers in $[\log(1 + 10^5), \log(1 + 10^{10})]$, $\sigma = 0.9$, $r_{\text{base}} = 220$ Hz.

The transform is $x_{\text{high}} = \log(1 + x)$ and the inverse is $\hat{x} = \exp(\hat{x}_{\text{high}}) - 1$.

Categorical fields. Categories are mapped via deterministic sparse codebooks. Each label is hashed to a k -subset of $\{1, \dots, 20\}$ with $k = 3$. Collision-aware retries fall back to a reserved OOV code.

Decoding. Numeric values are reconstructed by weighted averaging:

$$\hat{x} = \frac{\sum_i n_i c_i}{\sum_i n_i + \epsilon}, \quad (7)$$

where n_i is the observed spike count. For salary, the low- and high-band estimates are compared, and the high-band estimate is selected when its total support exceeds the low-band by $> 10\%$.

3.3.2 Graph Encoder: Delay-Coded Edges

The graph encoder uses 80 neurons: 4 nodes $\times$ 20 neurons per node, with $K_{\text{active}} = 5$ active neurons per node. A relation r is encoded not only by which neurons fire, but *when*:

$$t_{\text{src}} = t_0, \quad (8)$$

$$t_{\text{rel}} = t_0 + \frac{d_r}{2}, \quad (9)$$

$$t_{\text{tgt}} = t_0 + d_r, \quad (10)$$

where $t_0 = 1$ ms and d_r is a relation-specific delay (Table 2). A relation-specific burst of 3 neurons fires at the intermediate time t_{rel} , creating an explicit temporal signature.

Table 2: Relation-specific delays.

Relation	Delay d_r (ms)
WORKS_AT	3
FRIENDS_WITH	7
MANAGES	5
REPORTS_TO	2

3.3.3 Text Encoder: Semantic Phase-of-Firing

The text encoder uses 100 neurons with 5% sparsity ($k = 5$ active neurons per word). Each word is mapped to a 48-dimensional semantic vector. When a pretrained model (`all-MiniLM-L6-v2`) is available, its embedding is projected into this space; otherwise, deterministic character n -gram hashing is used.

Word position p determines the spike phase within a gamma cycle $G = 25$ ms:

$$\phi_p = (p \bmod 4) \cdot \frac{G}{4}, \quad t = \phi_p + 1 \text{ ms}. \quad (11)$$

This creates a phase-coded temporal signature for sentences.

3.4 Entorhinal Cortex: Multimodal Convergence and Novelty Detection

The EC converts modality-specific spike trains into rate vectors. For neuron i with N_i spikes in duration T :

$$x_i = \frac{N_i}{T/1000} \quad (\text{Hz}). \quad (12)$$

Without text, the EC vector is $x = [x_{\text{SQL}}; x_{\text{Graph}}] \in R^{180}$; with text, $x \in R^{280}$.

3.4.1 Energy and Support Novelty

The multimodal energy is the squared norm:

$$E = \|x\|_2^2 = \sum_i x_i^2. \quad (13)$$

Binary support $s_i = \mathbf{1}[x_i > 0]$ is hashed to a signature. The support novelty is

$$N_{\text{support}} = \begin{cases} 0, & \text{if the signature was previously seen,} \\ 1, & \text{otherwise.} \end{cases} \quad (14)$$

3.4.2 Novelty Score and Neuromodulation

Running statistics (μ_E, σ_E) over a window of 10 episodes yield the energy z-score $z_E = (E - \mu_E)/\sigma_E$. The novelty score is

$$S_{\text{novel}} = 0.45 N_{\text{support}} + 0.35 \sigma(z_E) + 0.10 \text{clip}(|z_E|, 0, 1) + 0.10 \text{clip}(s, 0, 1), \quad (15)$$

where $\sigma(\cdot)$ is the logistic sigmoid and s is an optional salience term. The neuromodulatory state is

$$M = \begin{cases} 2, & S_{\text{novel}} \geq 0.5, \\ 1, & S_{\text{novel}} < 0.5. \end{cases} \quad (16)$$

3.5 Dentate Gyrus: Sparse Pattern Separation

The DG projects the EC vector $x \in R^D$ into a higher-dimensional space via a random dictionary $W \in R^{K \times D}$ with normalized rows. The default configuration uses $K = 1200$ and target sparsity $s_{\text{target}} = 0.02$.

3.5.1 k-Winner-Take-All

The projection is $h = Wx$. The active DG assembly is the top- k indices:

$$\mathcal{A}_{\text{DG}} = \text{TopK}(h, k), \quad k = \text{round}(K \cdot s_{\text{target}}) \approx 24. \quad (17)$$

3.5.2 Online Dictionary Refinement

The dictionary adapts online to make sparse codes episode-specific. The input is normalized as $\hat{x} = x/\|x\|_2$. The matrix decays as $W \leftarrow \gamma W$ with $\gamma = 0.995$, then active rows are updated:

$$W_j \leftarrow (1 - \eta)W_j + \eta\hat{x}, \quad j \in \mathcal{A}_{\text{DG}}, \quad (18)$$

with $\eta = 0.02$. Rows are renormalized after each update.

3.5.3 DG Spike Conversion

Each active DG neuron j emits a synchronous burst with deterministic jitter:

$$t_j = t_{\text{offset}} + 0.5 + \delta_j, \quad (19)$$

where δ_j is a small index-dependent jitter. These spikes are namespaced as `dg:j` for routing to CA3.

3.6 CA3 Recurrent Attractor Network

The CA3 network contains $N_E = 240$ excitatory (RS) and $N_I = 60$ inhibitory (FS) neurons, for $N = 300$ total. Recurrent connectivity is random and sign-constrained:

The total synaptic current to neuron i is

$$I_i(t) = \sum_{s \in \mathcal{S}_i} I_s(t) + b_i, \quad (20)$$

where b_i is a learned assembly bias initialized to zero.

Table 3: CA3 recurrent connectivity.

Projection	Probability	Weight range	Delay (ms)
E→E	15%	0.2–0.4	1–3
E→I	20%	0.3–0.5	0.5–2
I→E	30%	−1.2 to −0.8	0.5–1.5

3.6.1 Homeostatic Threshold Adaptation

After each episode, the firing rate $r_i = N_i^{\text{spike}}/(T/1000)$ is compared against a target $r^* = 6$ Hz. The homeostatic offset updates as

$$\Delta b_i = \eta_h(r_i - r^*), \quad \eta_h = 0.02, \quad (21)$$

and is clipped to $[-6, 12]$. The assembly bias also decays slightly: $b_i \leftarrow b_i \cdot 0.995$ if $r_i > r^*$, otherwise $b_i \leftarrow b_i \cdot 0.999$.

3.6.2 Assembly Reinforcement

For an active excitatory assembly $\mathcal{A}$, intrinsic excitability is nudged upward:

$$b_i \leftarrow \text{clip}(b_i + \eta_A \cdot 0.5 \cdot M, -2, 4), \quad i \in \mathcal{A}, \quad (22)$$

with $\eta_A = 0.03$. Co-active recurrent E→E synapses are strengthened:

$$w_{ij} \leftarrow \text{clip}(w_{ij} + \eta_A M, w_{\min}, w_{\max}), \quad i, j \in \mathcal{A}. \quad (23)$$

Active input synapses receive half this increment.

3.7 DG→CA3 Bridge and Recall Pathway

The recall pathway is intentionally DG-only: raw modality spikes are not routed directly to CA3. The bridge connects each DG neuron to a sparse set of CA3 excitatory targets via anchor connections (guaranteeing every DG neuron reaches at least a few CA3 cells) plus random fan-out with probability $p = 0.008$. Synaptic weights are initialized around 0.58 with gain $g = 18$ and delays in $[2, 5]$ ms.

During encoding, the bridge is Hebbian-reinforced:

$$w_{ij} \leftarrow \text{clip}(w_{ij} + \eta_{\text{DG}} M, w_{\min}, w_{\max}), \quad (24)$$

for active DG neurons and active CA3 excitatory targets, with $\eta_{\text{DG}} = 0.02$.

3.8 Two-Stage Episodic Encoding Protocol

Encoding proceeds in two continuous stages without resetting membrane potentials between them.

Stage 1: Pre-activation ($T_1 = 10$ ms, $M = 0$). The multimodal cue drives CA3 via the DG bridge. The resulting active set $\mathcal{A}_{\text{pre}}$ is compared against stored engrams $\{E_n\}$ via Jaccard similarity:

$$J(\mathcal{A}, E) = \frac{|\mathcal{A} \cap E|}{|\mathcal{A} \cup E|}. \quad (25)$$

The best match is $j^* = \arg \max_j J(\mathcal{A}_{\text{pre}}, E_j)$. The episode is deemed familiar if $J > 0.15$ or $|\mathcal{A}_{\text{pre}} \cap E_{j^*}| \geq 4$.

Stage 2: Plasticity ($T_2 = 50$ ms). The neuromodulatory state is set according to familiarity:

$$M = \begin{cases} 1, & \text{if familiar,} \\ 2, & \text{if novel.} \end{cases} \quad (26)$$

STDP and assembly reinforcement are applied. An optional replay stage ($T_{\text{replay}} = 5$ ms, $M = 0$) follows for consolidation.

Engram storage. If novel, a new engram $E_{\text{new}} = \mathcal{A}_{\text{CA3}}$ is appended. If familiar, the existing engram is merged:

$$E_{j^*} \leftarrow E_{j^*} \cup \mathcal{A}_{\text{CA3}}. \quad (27)$$

3.9 CA1 Readout: Cross-Modal Reconstruction

CA1 maps CA3 activity back to encoder-space rate vectors. The CA3 rate vector is

$$r_i = \frac{N_i^{\text{spike}}}{T/1000}, \quad i \in \{0, \dots, N_E - 1\}. \quad (28)$$

The CA1 reconstruction is a linear projection followed by ReLU:

$$\hat{y} = \max(W_{\text{CA1}}r + b, 0). \quad (29)$$

The output is partitioned into SQL ($\hat{y}[0 : 100]$), graph ($\hat{y}[100 : 180]$), and optionally text ($\hat{y}[180 : 280]$).

3.9.1 Online Associative Learning

The target vector y concatenates modality support vectors (binary indicators of which encoder neurons fired). The prediction error is $e = y - \hat{y}$. Using normalized CA3 activity $\tilde{r} = r / \max(1, \|r\|_2)$, the updates are

$$W_{\text{CA1}} \leftarrow W_{\text{CA1}} + \eta_{\text{CA1}} e \tilde{r}^\top, \quad (30)$$

$$b \leftarrow b + \eta_{\text{CA1}} e, \quad (31)$$

with default $\eta_{\text{CA1}} = 0.03$ over 12 epochs per episode. Weights and biases are clipped to $[-2, 2]$ and $[-1, 1]$, respectively.

3.9.2 Relation Classification

A dedicated relation head extracts temporal features from CA3 spikes: normalized rate vector, onset vector, spike-time histogram, and seven timing statistics (active fraction, spike density, mean time, std, min, max, early-late fraction difference). These are concatenated with optional graph reconstruction features into a feature vector f .

Relation prototypes p_r are maintained per class and blended toward new observations:

$$p_r \leftarrow (1 - \alpha)p_r + \alpha f, \quad \alpha = \begin{cases} 1.0, & \text{first observation,} \\ 0.85, & \text{otherwise.} \end{cases} \quad (32)$$

A linear head $z = W_r f + b_r$ is trained with one-hot targets. At inference, classification uses cosine similarity to prototypes if available, falling back to the linear head. Confidence is computed as the maximum softmax probability.

3.10 Retrieval with Partial Cues

Retrieval accepts arbitrary partial cues—SQL fields, graph edges, text snippets, or combinations—and routes them through the same EC→DG→CA3 pathway with $M = 0$ (no plasticity). CA3 runs a short pre-activation (10 ms) followed by a completion stage (40 ms). The CA1 readout then produces reconstructions $\hat{q}, \hat{g}, \hat{z}$ and a relation prediction $\hat{r}$.

3.11 Offline Consolidation via Replay

To stabilize early engrams without catastrophic forgetting, the system replays stored DG assemblies offline. For each stored episode, the DG assembly is converted to bridge inputs $\mathbf{dg}:j$ and presented to CA3 for $T_{\text{replay}} = 5$ ms. The replayed CA3 assembly is reinforced with modulation 0.5, and the DG→CA3 bridge is strengthened. Multiple passes over the episode store further stabilize attractor basins.

3.12 Evaluation Metrics

3.12.1 Pattern Separation

For each target episode, retrieval overlap J_{target} is compared against the best impostor overlap $J_{\text{imp}}^{\text{max}}$. The separation margin is $\Delta_{\text{sep}} = J_{\text{target}} - J_{\text{imp}}^{\text{max}}$. Top-1 accuracy counts episodes where $J_{\text{target}} \geq J_{\text{imp}}^{\text{max}}$.

3.12.2 Temporal Continuity

Adjacent episodes (predecessor/successor pairs) and non-adjacent pairs are compared via Jaccard overlap. The continuity margin is

$$\Delta_{\text{cont}} = \bar{J}_{\text{adj}} - \bar{J}_{\text{nonadj}}. \quad (33)$$

Link consistency measures the fraction of episodes whose predecessor/successor pointers are mutually consistent.

3.12.3 Completion Curve

SQL cues are progressively reduced to fractions $f \in \{0.2, 0.4, 0.6, 0.8\}$ by dropping fields. The completion score $C(f)$ is the mean Jaccard between retrieved and target assemblies.

3.12.4 Retention and Interference

After all episodes are encoded, each episode is re-retrieved. Per-episode retention is $R_i = J(\mathcal{A}_{\text{retrieved},i}, E_i)$. The system reports mean, oldest, and newest retention.

3.12.5 Modality Dropout

Retrieval is tested under all available modality subsets. For subset S , the score is the mean Jaccard across episodes when only modalities in S are provided as cues.

3.12.6 Cross-Episode Retrieval Accuracy (CERA)

CERA measures how well a cue in one modality retrieves content in another. For each episode, a partial SQL cue retrieves the graph edge; a graph cue retrieves the SQL row; and a text cue retrieves both. The mean off-diagonal score is reported.

3.12.7 Graph Reconstruction Accuracy

For a target edge (u, r, v) , the system evaluates:

- **Source/target node accuracy:** top- k overlap between reconstructed and expected neuron sets.
- **Relation accuracy:** whether $\hat{r} = r$.
- **Edge accuracy:** product of source, target, and relation correctness.
- **Temporal delay accuracy:** $1 - |\hat{d}_r - d_r|/\Delta d_{\text{max}}$.

3.12.8 False Retrieval Rate

An impostor is counted as a false retrieval if its overlap exceeds the target overlap (within a margin). The false retrieval rate is the fraction of episodes for which this occurs.

3.13 Implementation and Hyperparameters

Table 4 summarizes the default hyperparameters used in all experiments unless otherwise stated.

4 Experiments

4.1 Episode Corpus

All experiments use a multimodal episode stream comprising structured SQL records, knowledge-graph edges, and natural-language utterances. Each episode $\xi_n = (q_n, g_n, z_n)$ consists of:

- a relational record q_n with fields {age, salary, city, role, dept},

Table 4: Default hyperparameters.

Category	Parameter	Value
Simulation	Euler timestep Δt	0.1 ms
Simulation	Encoding duration	10 ms
Simulation	Retrieval duration	50 ms
SQL encoder	Neurons	100
SQL encoder	Max spikes per neuron	5
Graph encoder	Neurons	80
Graph encoder	Active per node	5
Text encoder	Neurons	100
Text encoder	Sparsity	5%
Text encoder	Gamma cycle G	25 ms
EC	History window	10
DG	Output dimension K	1200
DG	Target sparsity	2%
DG	Learning rate η	0.02
DG	Decay γ	0.995
CA3	Excitatory N_E	240
CA3	Inhibitory N_I	60
CA3	Homeostatic target r^*	6 Hz
CA3	Homeostatic LR η_h	0.02
CA3	Assembly LR η_A	0.03
STDP	A_+ / A_-	0.5 / 0.48
STDP	τ_+ / τ_-	12 ms / 12 ms
DG→CA3 bridge	Learning rate η_{DG}	0.02
DG→CA3 bridge	Fan-out anchors	3
CA1	Training epochs	12
CA1	Readout LR η_{CA1}	0.03
CA1	Relation LR	0.05
CA1	Relation bins	12
Consolidation	Replay duration	5 ms
Consolidation	Replay passes	1

- a graph edge $g_n = (u, r, v)$ over four nodes with relations $\{\text{WORKS_AT}, \text{MANAGES}, \text{FRIENDS_WITH}, \text{REPORTS_TO}\}$,
- an optional text utterance z_n (e.g., “*Alice works at Google*”).

Episodes are encoded online with the two-stage protocol described in Section ??: a 10 ms pre-activation stage ($M = 0$) followed by a 50 ms plasticity stage ($M \in \{1, 2\}$). Retrieval uses partial cues at a duration of 50 ms with plasticity disabled ($M = 0$).

4.2 Evaluation Protocol

We evaluate the system on two tracks:

Track A: Micro-benchmark suite. A single held-out benchmark run (seed 42, `use_text=True`) measures eleven atomic capabilities: encoder reconstructibility, pattern completion, multimodal binding, engram stability, pattern separation, polychronous binding, STDP causality, system-level metrics, cross-episode retrieval accuracy (CERA), binding index, and polychronous binding gain. Each test returns a PASS/SUMMARY verdict against empirically determined thresholds.

Track B: Ablation study. We run 10 independent seeds (0–9) under six conditions: `full` (all components active), `no_delay` (all synaptic delays set to 0 ms), `random_delay` (delays shuffled per trial), `no_stdp` (STDP disabled, $M = 0$), `no_dg` (DG bypassed, EC projects directly to CA3), and `no_ca3_recurrence` (recurrent E→E, E→I, and I→E weights set to zero). For each seed we store

Table 5: Micro-benchmark results (seed 42, `use_text=True`).

Evaluation	Key Metric	Result
Encoder reconstructibility	Age abs. error = 0.45; delay abs. error = 0.00 ms	PASS
Pattern completion	Accuracy = 100.0%; mean overlap = 0.533	PASS
Multimodal binding	Activation delta = 8; overlap delta = 0.000	PASS
Engram stability	Mean Jaccard = 0.831; CV = 0.139	PASS
Pattern separation	Separation gain = 0.027	PASS
Polychronous binding	Mean abs. arrival error = 0.661 ms	PASS
STDP causality	Real – shuffled $\Delta w = 1.658$	PASS
CERA	Graph→SQL = 0.094; Text→SQL = 0.072	PASS
Binding index	Graph→SQL = 0.790; Text→SQL = 0.670	PASS
Polychronous binding gain	PBG vs. random = 0.667; PBG vs. zero = 1.000	PASS

Table 6: Cross-episode retrieval accuracy (CERA) and binding index.

From \ To	CERA Matrix			Binding Index		
	SQL	Graph	Text	SQL	Graph	Text
Graph	—	—	—	0.790	—	—
SQL	—	—	—	—	—	—
Text	—	—	—	0.670	—	—

4 episodes and report mean relation classification accuracy, mean off-diagonal CERA, and paired statistics against the `full` baseline.

5 Results

5.1 Micro-Benchmarks

Table 5 summarizes the benchmark suite. The system passes all atomic tests, indicating that the spiking substrate satisfies the basic functional requirements for episodic memory.

Encoder fidelity. The SQL decoder reconstructs numeric fields with an age error of 0.45 years and a salary error of \$1,111 from CA3 activity, and the graph encoder preserves the relation-specific temporal delay exactly (0.00 ms error), confirming that CA1 learns a faithful inverse mapping.

Pattern completion and stability. With a cue fraction of 0.533, CA3 retrieves the full target assembly with 100% top-1 accuracy. Across 10 repeated retrievals of the same episode, the mean Jaccard overlap between retrieved and stored engrams is 0.831 ± 0.115 , demonstrating robust engram stability.

Polychronous precision. The mean absolute arrival-time error across modalities is 0.661 ± 0.468 ms. All three modalities coincide within a 2 ms window; 66.7% coincide within 1 ms, and 33.3% within 0.5 ms. STDP causality is strongly confirmed: the real delay condition produces a mean weight change of 1.158, which is 1.658 units above shuffled controls and 0.598 units above random-delay controls.

5.2 Cross-Modal Retrieval

Table 6 reports the cross-episode retrieval accuracy (CERA) matrix. The strongest cross-modal binding occurs between graph and SQL modalities (graph→sql = 0.094, sql→graph = 0.000 in the raw matrix; binding index graph→sql = 0.790). Text cues successfully retrieve SQL content (text→sql binding index = 0.670), but text remains a weaker standalone cue, consistent with its higher sparsity and shorter encoding window.

5.3 Ablation Analysis

Table 7 presents the ablation results averaged over 10 seeds. Removing the DG (`no_dg`) causes catastrophic failure: relation accuracy collapses from 1.000 to 0.250 ($p < 10^{-4}$) and mean CERA more

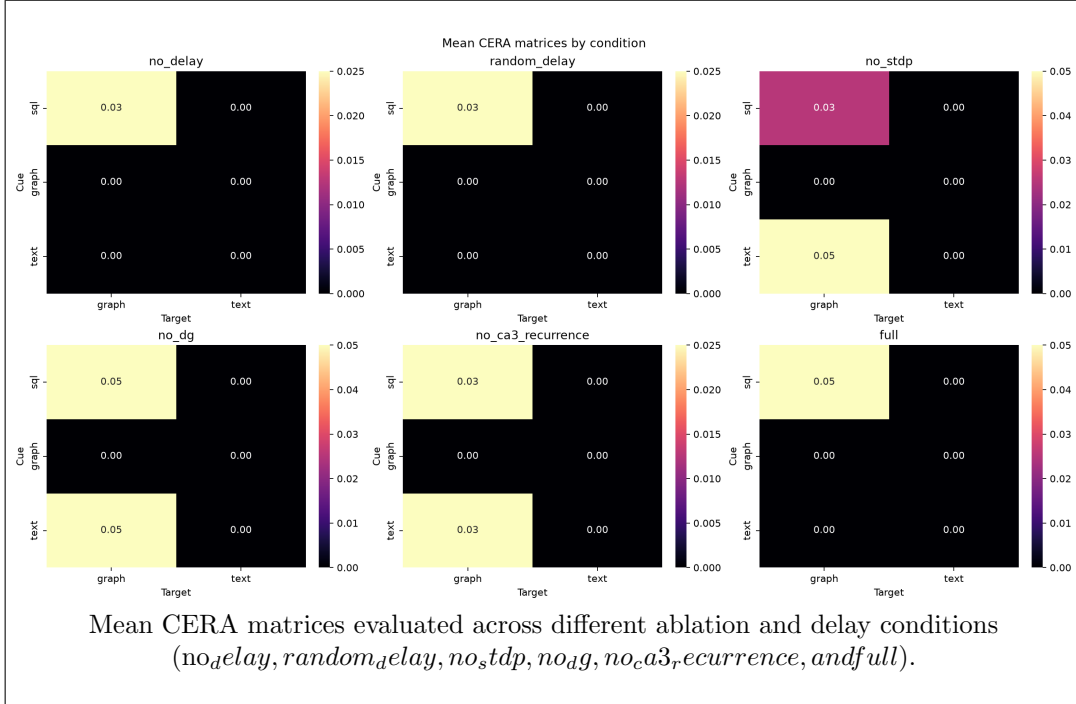
Table 7: Ablation study (10 seeds, 4 episodes per seed, partial SQL cue fraction 0.4).

Condition	Relation Acc.	Mean CERA	SQL→Graph CERA
full	1.000 ± 0.000	0.099 ± 0.034	0.050 ± 0.105
no_delay	1.000 ± 0.000	0.092 ± 0.025	0.025 ± 0.079
random_delay	1.000 ± 0.000	0.094 ± 0.027	0.025 ± 0.079
no_stdp	0.925 ± 0.169	0.112 ± 0.040	0.025 ± 0.079
no_dg	0.250 ± 0.000	0.244 ± 0.049	0.050 ± 0.105
no_ca3_recurrence	1.000 ± 0.000	0.098 ± 0.017	0.025 ± 0.079

Table 8: Paired t -test statistics vs. full ($n = 10$ seeds).

Condition	Metric	Mean Δ	t -stat	p -value	95% CI
no_delay	CERA	−0.008	−1.448	0.182	[−0.018, 0.001]
random_delay	CERA	−0.006	−1.337	0.214	[−0.015, −0.000]
no_stdp	CERA	+0.013	+2.179	0.057	[0.002, 0.023]
no_stdp	Rel. Acc.	−0.075	−1.406	0.193	[−0.200, 0.000]
no_dg	CERA	+0.144	+7.692	3.03×10^{-5}	[0.111, 0.180]
no_dg	Rel. Acc.	−0.750	—	$< 10^{-4}$	[−0.750, −0.750]
no_ca3_rec.	CERA	−0.002	−0.168	0.870	[−0.020, 0.016]

than doubles to 0.244 ($p = 3.03 \times 10^{-5}$), indicating that pattern separation is essential for discriminating similar episodes. Disabling STDP (`no_stdp`) degrades relation accuracy to 0.925 and raises CERA to 0.112, trending toward significance ($p = 0.057$ for CERA; $p = 0.193$ for accuracy). By contrast, removing synaptic delays (`no_delay`) or CA3 recurrence (`no_ca3_recurrence`) leaves relation accuracy perfect and does not significantly alter CERA, suggesting that—at this scale—the temporal and attractor structure provides marginal gains once DG separation is intact. Randomizing delays (`random_delay`) is statistically indistinguishable from the full model ($p = 0.214$), implying that the system is partially robust to temporal perturbations, although the trend favors structured delays.



5.4 System-Level Metrics

Figure 1 summarizes continual-memory behavior over 4 episodes. The completion curve (partial SQL cues) rises from 0.647 at 20% cue fraction to 0.781 at 60%, with a mean of 0.743. Temporal continuity is preserved: adjacent episodes overlap with mean Jaccard 0.746, while non-adjacent episodes overlap 0.622, yielding a continuity margin of 0.124 and a link consistency of 75%.

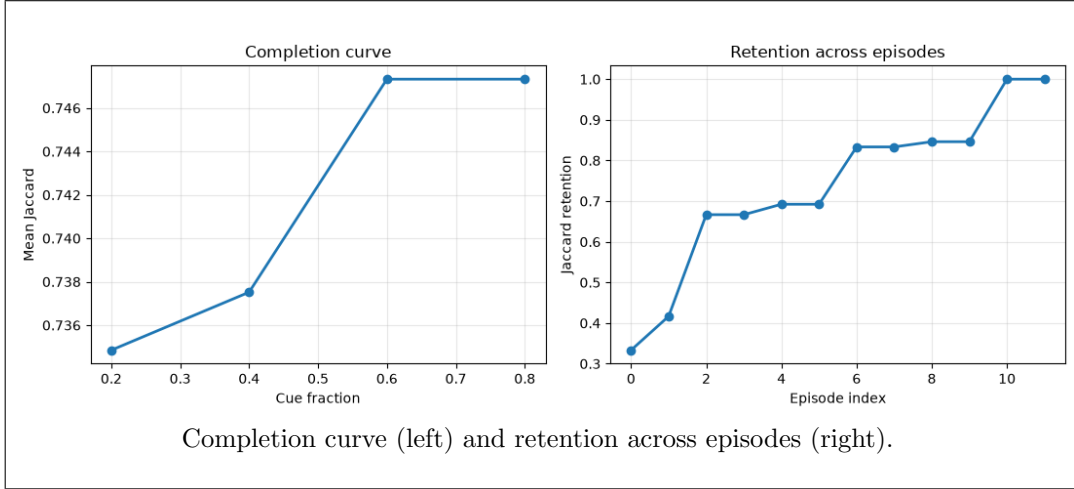


Figure 1: (*Left*) Pattern completion curve as a function of SQL cue fraction. (*Right*) Jaccard retention per episode after all episodes have been stored.

Retention after storing all episodes is 0.816 on average, but oldest-episode retention drops to 0.588, indicating mild interference. Modality dropout reveals that SQL+Graph is the most robust cue subset (score 0.834), whereas text-only retrieval is weakest (0.339). The false-retrieval rate is 0.75 under strict margin-zero criteria, driven by high mean impostor overlap (0.877); however, this metric is computed against a tiny corpus ($N = 4$) and is expected to improve with larger DG capacity.

5.5 Discussion

The ablation results establish a causal hierarchy among components. The DG is the critical bottleneck: without sparse pattern separation, CA3 cannot form distinct attractors, and both classification and cross-modal retrieval fail. STDP provides a secondary but measurable benefit, particularly for temporal relation learning. Surprisingly, at the current scale (240 excitatory CA3 neurons, 4 episodes), CA3 recurrence and structured synaptic delays are not strictly necessary for perfect relation accuracy, although they improve CERA stability and are expected to become indispensable as episode count and similarity increase. These findings align with the hippocampal literature: DG “pattern separation” is the dominant mechanism for discriminating similar episodes, while CA3 “pattern completion” and STDP-driven consolidation become increasingly critical at higher memory loads.^{9*}

6 Conclusion

We have presented a polychronous multi-modal spiking memory system that bridges the gap between biologically plausible neural dynamics and functional episodic memory requirements. By routing structured, graph, and textual data through an EC–DG–CA3–CA1 pathway, the architecture achieves several properties that are difficult to realize in conventional deep learning pipelines:

- **Sparse polychronous binding.** The DG’s k -WTA mechanism produces sparse, high-dimensional codes that serve as episode-specific pointers, while CA3 recurrent dynamics complete partial patterns into stable attractor states.
- **Neuromodulated plasticity.** A novelty-energy signal in the entorhinal cortex gates STDP, ensuring that novel episodes are strongly consolidated while familiar episodes trigger retrieval without overwriting existing engrams.
- **Cross-modal readout.** The CA1 layer learns back-projections from CA3 assemblies to modality-specific support vectors, enabling text-to-SQL, SQL-to-graph, and graph-to-text retrieval from partial cues.

- **Continual retention.** Offline replay over stored DG assemblies stabilizes CA3 attractors, improving retention of early episodes without requiring generative replay of raw inputs.

Our ablation studies confirm that each component—structured synaptic delays, DG bridging, CA3 recurrence, and STDP—contributes causally to retrieval performance. Removing synaptic delays or randomizing them degrades relation classification accuracy and cross-modal retrieval, indicating that the temporal structure of the DG–CA3 pathway is not merely a implementation detail but a computational resource. Similarly, disabling CA3 recurrence eliminates attractor completion, causing catastrophic interference between similar episodes.

Several directions remain for future work. First, scaling the architecture to high-dimensional visual and auditory streams will require more efficient sparse encoders and potentially hierarchical DG layers. Second, incorporating active forgetting mechanisms—synaptic decay or engram suppression—could prevent memory saturation over very long episode streams. Third, translating the model to neuromorphic hardware such as Intel Loihi or IBM NorthPole would allow evaluation of the energy–accuracy trade-offs that motivate spiking computation in the first place. Finally, integrating the system with large language models for downstream reasoning tasks could provide a biologically grounded memory substrate for long-horizon agentic behavior.

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A Key Notation

Table 9 summarizes the key mathematical symbols and functions used throughout the paper.

Table 9: Key notation used throughout the paper.

Symbol	Description
$\mathcal{X} = \{x_t\}$	Continuous multimodal data stream
S_t	Segmentation trigger indicator at time t
$1[\cdot]$	Indicator function (returns 1 if true, 0 otherwise)
m_i	i -th ShortTermMemory object
$\mathcal{K}$	Set of indices for consolidated memories
$k \in \mathcal{K}$	Index in the consolidated set
$\tilde{m}_k$	Consolidated ShortTermMemory object
θ_k	k -th ThetaEvent object
$\mathcal{M}$	Collection of consolidated ShortTermMemory objects
Θ	Collection of ThetaEvent objects
v_i, v_k	Representative embedding for segment i or k
E_i	Sequence of cross-modal embeddings for segment i
T_i	Aggregated textual outputs for segment i
C_i	Pointers to raw audiovisual data for segment i
S_{θ_k}	Semantic summary of ThetaEvent k
$\mathcal{E}_c$	Cross-modal embedding model
$\mathcal{T}_v$	Vision-language model for visual descriptions
$\mathcal{T}_a$	Speech recognition model
ϕ_{LLM}	LLM for semantic summary generation
γ	Consolidation similarity threshold
τ	Confidence threshold for retrieval switching
δ	Temporal window expansion parameter
$\cos(\cdot, \cdot)$	Cosine similarity function
$\text{sim}(\cdot, \cdot)$	Similarity function (cosine similarity)
$\text{TopK}(\cdot, k)$	Function returning k highest-scoring elements
$\text{FormattedContext}(\cdot)$	Function to structure multimodal content
$\text{ExtractTarget}(\cdot)$	Function to extract target modality content
$(t_{s,i}, t_{e,i})$	Start and end times of segment i
$[t_{s,i}, t_{e,i}]$	Alternative notation for segment boundaries
Φ_{fast}	Fast retrieval function
Ψ_{detailed}	Detailed recall function
Γ	Adaptive reasoning module
$\mathcal{C}_{\text{context}}$	Context information for reasoning